

GMM-Based Anomaly Detection: Learning Beyond Gaussian Assumptions

Group - FortyTwo

Suhani
IIT Gandhinagar
Gandhinagar, Gujarat, India
24110358@iitgn.ac.in

Mateen Sharief
IIT Gandhinagar
Gandhinagar, Gujarat, India
24110197@iitgn.ac.in

Mayank Choudhary
IIT Gandhinagar
Gandhinagar, Gujarat, India
24110199@iitgn.ac.in

Lingala Hasini
IIT Gandhinagar
Gandhinagar, Gujarat, India
24110186@iitgn.ac.in

Abstract

This project extends the Deep Autoencoding Gaussian Mixture Model (DAGMM) for unsupervised anomaly detection by relaxing its Gaussian distributional assumptions. We evaluate three progressively expressive density-estimation strategies: (1) substituting Gaussian mixture components with Student- t , Laplace, and skew-normal families; (2) replacing the mixture with a RealNVP normalizing-flow density estimator (FlowDAGMM); and (3) combining both in a hybrid Non-Gaussian Mixture Flow (NGMF) with per-component flows over a Student- t base. A temperature-annealed estimation network removes the need for manual component-count selection. Experiments on KDD Cup 99, MIT-BIH Arrhythmia, and Thyroid—compared against Isolation Forest, One-Class SVM, VAE, and Deep SVDD—show that FlowDAGMM achieves the best performance among our methods (AUROC 0.998 on KDD99, 0.817 on MIT-BIH), while revealing that DAGMM’s limitations stem more from optimisation instability than from Gaussian assumptions alone.

CCS Concepts

• **Computing methodologies** → **Anomaly detection**; *Machine learning*.

Keywords

anomaly detection, Gaussian mixture models, non-Gaussian distributions, deep autoencoder, normalizing flows, Dirichlet Process, density estimation

1 Introduction

Anomaly detection—identifying data points that deviate significantly from the majority—is fundamental across fraud detection, network intrusion, industrial fault diagnosis, and healthcare monitoring. In high-dimensional settings the curse of dimensionality makes density estimation and threshold selection challenging.

The Deep Autoencoding Gaussian Mixture Model (DAGMM) [13] addresses this by coupling a deep autoencoder with a Gaussian Mixture Model (GMM) in an end-to-end trainable architecture. The autoencoder compresses input $\mathbf{x}$ to a low-dimensional code $\mathbf{z}_c$ and reconstructs $\mathbf{x}'$. The latent vector $\mathbf{z} = [\mathbf{z}_c \mid \text{rel-Euclid.} \mid \text{cos-sim}]$ is

then modelled by the GMM, whose negative log-likelihood serves as the anomaly score.

However, the Gaussian assumption may be inappropriate for real-world data exhibiting heavy tails, skewness, or multimodal sub-structure. This project systematically extends DAGMM along three directions of increasing expressiveness:

- (1) **Strategy 1 (Non-Gaussian Components):** Replace Gaussian mixture components with Student- t , Laplace, or skew-normal distributions.
- (2) **Strategy 2 (FlowDAGMM):** Replace the GMM entirely with a RealNVP normalizing-flow density estimator [2].
- (3) **Strategy 3 (NGMF):** Equip each mixture component with its own lightweight RealNVP flow over a Student- t base, combining mixture interpretability with flow flexibility.

We additionally incorporate a Dirichlet-Process-inspired estimation network with temperature annealing to learn the effective number of components automatically. We evaluate all methods on KDD Cup 99 [1], MIT-BIH Arrhythmia [7], and Thyroid, comparing against Isolation Forest [6], One-Class SVM [12], VAE [4], and Deep SVDD [11].

2 Related Work

DAGMM. Zong et al. [13] introduced DAGMM as an end-to-end deep model for unsupervised anomaly detection. A compression network produces a latent code augmented with reconstruction-error features; an estimation network predicts soft GMM assignments; and the sample energy

$$E(\mathbf{z}) = -\log \sum_k \phi_k \mathcal{N}(\mathbf{z}; \mu_k, \Sigma_k)$$

serves as the anomaly score.

Non-Gaussian Mixtures. Peel and McLachlan [9] showed that Student- t mixtures offer robustness to outliers due to heavier tails. Lin et al. [5] extended this to skew- t mixtures capable of modelling asymmetric clusters.

Normalizing Flows. Rezende and Mohamed [10] introduced normalizing flows for variational inference. RealNVP [2] and Glow [3] demonstrated that affine coupling layers yield tractable, expressive density estimators. Papamakarios et al. [8] provided masked autoregressive alternatives.

One-Class Methods. Deep SVDD [11] maps data to a hypersphere and scores anomalies by distance to the centre. Isolation Forest [6] isolates anomalies via random recursive partitioning. One-Class SVM [12] estimates the support of the data distribution via kernel methods.

3 Methodology

All strategies share a common deep autoencoder backbone and differ only in their density-estimation head.

3.1 Shared Architecture

The encoder maps an input $\mathbf{x} \in \mathbb{R}^d$ through fully-connected layers $d \rightarrow 60 \rightarrow 30 \rightarrow 10 \rightarrow d_z$ (with Tanh activations) to produce latent code $\mathbf{z}_c$. The decoder mirrors this. The augmented representation is:

$$\mathbf{z} = [\mathbf{z}_c \mid \frac{\|\mathbf{x} - \mathbf{x}'\|_2}{\|\mathbf{x}\|_2 + \epsilon} \mid \cos(\mathbf{x}, \mathbf{x}')] \in \mathbb{R}^{d_z+2}. \quad (1)$$

3.2 Baseline DAGMM

The estimation network predicts soft GMM assignments $\gamma = \text{softmax}(\text{MLP}(\mathbf{z}))$. Mixture parameters $\{\phi_k, \mu_k, \Sigma_k\}$ are computed from $(\mathbf{z}, \gamma)$ per batch, and the sample energy is:

$$E(\mathbf{z}) = -\log \sum_{k=1}^K \phi_k \mathcal{N}(\mathbf{z}; \mu_k, \Sigma_k). \quad (2)$$

The loss is $\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_1 \bar{E} + \lambda_2 \sum_k \|\text{diag}(\Sigma_k^{-1})\|_F$.

3.3 Strategy 1: Non-Gaussian Mixture Components

We replace each Gaussian component with one of three distributional families.

Student- t Mixture. Each component follows a multivariate Student- t with learnable degrees of freedom ν_k . Let $\Delta_k^2 = (\mathbf{z} - \mu_k)^\top \Sigma_k^{-1} (\mathbf{z} - \mu_k)$, then:

$$p_k(\mathbf{z}) = \frac{\Gamma(\frac{\nu_k+d}{2})}{\Gamma(\frac{\nu_k}{2}) (\nu_k \pi)^{d/2} |\Sigma_k|^{1/2}} \left(1 + \frac{\Delta_k^2}{\nu_k}\right)^{-\frac{\nu_k+d}{2}}. \quad (3)$$

Laplace Mixture. Components use a multivariate Laplace kernel:

$$p_k(\mathbf{z}) \propto \exp(-\sqrt{\Delta_k^2}). \quad (4)$$

Skew-Normal Mixture. A Gaussian density modulated by a learned skewness vector α_k :

$$p_k(\mathbf{z}) = 2 \mathcal{N}(\mathbf{z}; \mu_k, \Sigma_k) \Phi(\alpha_k^\top \Sigma_k^{-1/2} (\mathbf{z} - \mu_k)). \quad (5)$$

All energies are computed in log-space via logsumexp for numerical stability.

3.4 DP-Inspired Estimation Network

Standard softmax estimation tends to collapse all mass into one component. We replace it with temperature-annealed softmax:

$$\gamma_{n,k} = \frac{\exp(l_{n,k}/\tau)}{\sum_j \exp(l_{n,j}/\tau)}, \quad (6)$$

where τ is annealed from 2.0 to 0.5 over training (factor 0.95 per epoch). A KL penalty $\mathcal{L}_{\text{KL}} = \text{KL}(\tilde{\gamma} \parallel \text{Uniform}(K))$ encourages unused components to deactivate.

3.5 Strategy 2: FlowDAGMM (RealNVP)

We replace the GMM entirely with a RealNVP normalizing flow [2]. The flow $f_\theta : \mathbb{R}^{d_z+2} \rightarrow \mathbb{R}^{d_z+2}$ consists of 6 affine coupling layers (hidden dimension 16, alternating masks):

$$\log p(\mathbf{z}) = \log p_0(f_\theta(\mathbf{z})) + \log \left| \det \frac{\partial f_\theta}{\partial \mathbf{z}} \right|, \quad (7)$$

where $p_0 = \mathcal{N}(\mathbf{0}, \mathbf{I})$. The anomaly score is $-\log p(\mathbf{z})$. The loss is $\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_1 \cdot \text{NLL}$, trained with cosine-annealing learning rate.

3.6 Strategy 3: Non-Gaussian Mixture Flow (NGMF)

Our most expressive strategy combines mixture interpretability with flow flexibility:

$$p(\mathbf{z}) = \sum_{k=1}^K \phi_k p_k(f_{\theta_k}(\mathbf{z})) \cdot \left| \det \frac{\partial f_{\theta_k}}{\partial \mathbf{z}} \right|, \quad (8)$$

where each component has a lightweight 4-layer RealNVP flow (hidden 8) and a Student- t base distribution with learnable ν_k . The estimation network uses temperature-annealed softmax. The loss is $\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_1 \text{NLL} + \lambda_3 \mathcal{L}_{\text{KL}}$.

4 Experimental Setup

4.1 Datasets

- **KDD Cup 99** [1]: Network intrusion detection; 494,021 samples; 41 features (one-hot encoded to 118). Normal class: normal.; all other labels are anomalies.
- **MIT-BIH Arrhythmia** [7]: Cardiac ECG recordings. Normal class: type 'N'; all other beat types are anomalies. Features are the signal amplitudes.
- **Thyroid**: 3,772 samples, 6 features from the UCI repository. Standard medical anomaly benchmark.

All datasets are split 80/20 (train/test), and only normal samples are used for training (semi-supervised protocol). Features are standardised with StandardScaler.

4.2 Baselines

- **Isolation Forest** [6]: 200 estimators, auto contamination.
- **One-Class SVM** [12]: RBF kernel, $\nu = 0.1$, subsampled to 10K training points.
- **VAE** [4]: Encoder $60 \rightarrow 30$, latent dim 8, MSE reconstruction anomaly score.
- **Deep SVDD** [11]: Encoder $60 \rightarrow 30 \rightarrow 32$, 20-epoch autoencoder pretraining + 50-epoch SVDD training.

4.3 Training Configuration

All deep models use Adam (lr 10^{-4}), batch size 256, 50 epochs, and gradient clipping at norm 5. DAGMM and Strategy 1 use $\lambda_1=0.1$, $\lambda_2=0.005$, $\lambda_3=0.01$, with $K=4$ components and $d_z=1$. FlowDAGMM uses 6 flow layers (hidden 16). NGMF uses 4 per-component flow layers (hidden 8).

4.4 Evaluation Metrics

We report AUROC and the best-threshold F1 score (with corresponding Precision and Recall), scanning 200 percentile thresholds between 1 and 99.

5 Results and Analysis

5.1 Overall Comparison

Table 1 presents the complete results across all datasets.

5.2 KDD Cup 99

On this large-scale network intrusion dataset, Deep SVDD achieves the highest AUROC (0.999), closely followed by FlowDAGMM (0.998) and NGMF (0.995). Classical baselines also perform strongly: Isolation Forest (0.987) and One-Class SVM (0.955). Baseline DAGMM fails catastrophically (AUROC 0.093), attributable to training instability when GMM parameters are estimated per-batch.

Among Strategy 1 variants, skew-normal achieves the best AUROC (0.925) but with extremely low recall (0.032), indicating over-conservative scoring. FlowDAGMM’s success suggests that the bottleneck is not the Gaussian assumption per se, but rather the difficulty of jointly optimising covariance estimation and reconstruction.

5.3 MIT-BIH Arrhythmia

This cardiac anomaly dataset is the most challenging. Deep SVDD leads (0.844), with FlowDAGMM second (0.817) and One-Class SVM third (0.812). NGMF drops to 0.703, suggesting per-component flow optimisation struggles when anomaly structure is subtle. Strategy 1 variants all fall below 0.65, with Laplace achieving the best among them. Baseline DAGMM achieves near-chance AUROC (0.513).

5.4 Thyroid

Isolation Forest dominates this small tabular dataset (AUROC 0.994, F1 0.790), highlighting that tree-based methods excel when dimensionality is low. FlowDAGMM and Deep SVDD achieve comparable AUROC (~0.933) but much lower F1 scores. Strategy 1 variants show moderate improvement over baseline DAGMM, with skew-normal reaching 0.834 AUROC.

5.5 Visualisation Analysis

Figure 1 shows comparative bar charts across all models. Figure 3 presents t-SNE embeddings of the latent representations; FlowDAGMM produces well-separated normal/anomaly clusters on KDD99 but less distinct separation on MIT-BIH. Figure 4 shows kernel density estimates of anomaly score distributions, where FlowDAGMM achieves the best separation between normal and anomalous samples.

5.6 Key Findings

- (1) **FlowDAGMM is the best proposed method**, achieving competitive AUROC with Deep SVDD and outperforming all DAGMM variants. This validates the hypothesis that a flow-based density improves anomaly detection.

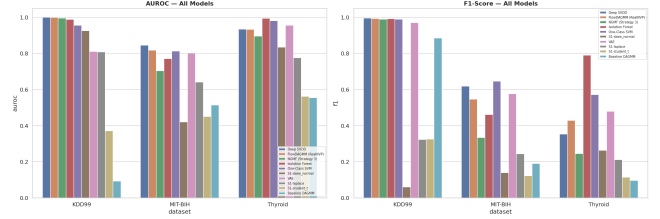


Figure 1: AUROC and F1 comparison across all models and datasets.

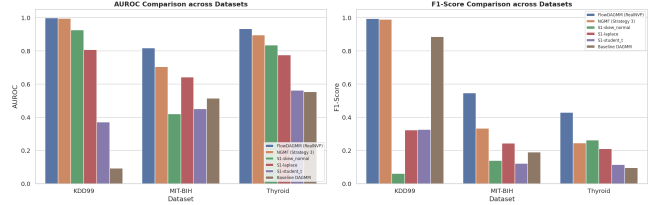


Figure 2: Comparison of the three proposed strategies across datasets.

- (2) **Strategy 1 largely underperforms.** Despite theoretical appeal, the Student- t , Laplace, and skew-normal variants struggle with optimisation; the additional parameters interact poorly with per-batch GMM estimation.
- (3) **NGMF does not outperform FlowDAGMM.** The per-component flow architecture adds complexity without commensurate gains, likely because the small latent dimension does not benefit from multiple local flows.
- (4) **Baseline DAGMM is unstable.** AUROC below 0.10 on KDD99 confirms known issues with energy-based training when per-batch GMM statistics are noisy. Global-GMM evaluation (computing ϕ , μ , Σ over the full training set) only partially addresses this.
- (5) **Dataset characteristics matter.** Isolation Forest dominates on low-dimensional Thyroid; Deep SVDD excels on higher-dimensional data. No single method is universally best.

6 Conclusion

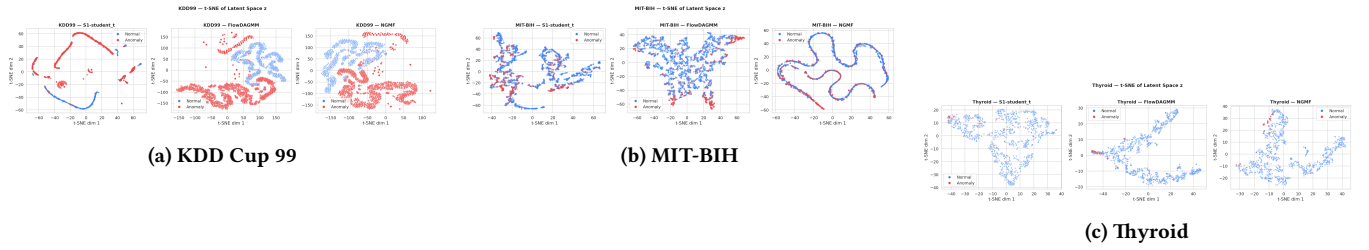
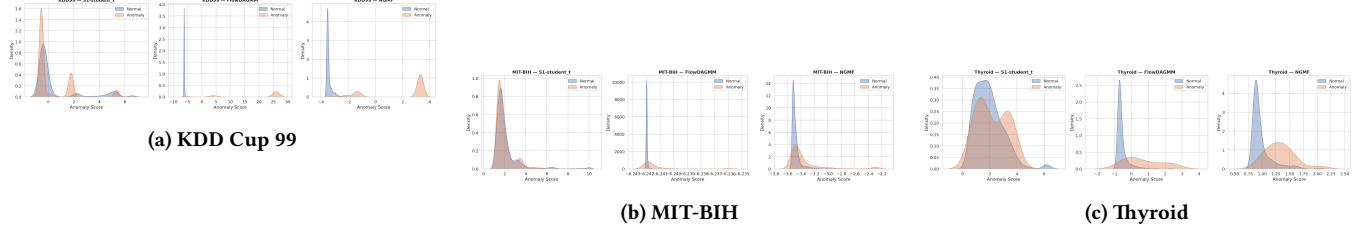
We presented a systematic study extending DAGMM beyond Gaussian assumptions through three strategies of increasing expressiveness. Our experiments on three diverse benchmarks reveal that FlowDAGMM—replacing the GMM with a RealNVP normalizing flow—is the most effective extension, achieving near-state-of-the-art AUROC while remaining end-to-end trainable.

Contrary to our initial hypothesis, replacing Gaussian components with heavier-tailed distributions (Strategy 1) does not yield consistent improvements. This suggests that DAGMM’s primary limitation is optimisation difficulty rather than the Gaussian assumption. The NGMF hybrid (Strategy 3), while theoretically the most expressive, faces optimisation challenges that prevent it from surpassing the simpler FlowDAGMM.

Future Work. Promising directions include: (1) scaling with more coupling layers or Glow-style 1×1 convolutions; (2) applying

Table 1: Anomaly detection results across three benchmark datasets. Best result per dataset in bold; best proposed method underlined.

Type	Model	KDD Cup 99				MIT-BIH				Thyroid			
		AUC	F1	Prec	Rec	AUC	F1	Prec	Rec	AUC	F1	Prec	Rec
Baselines	Deep SVDD	.999	.995	.999	.991	.844	.618	.654	.586	.933	.353	.245	.632
	Isolation Forest	.987	.991	.995	.987	.770	.461	.488	.437	.994	.790	.790	.790
	One-Class SVM	.955	.989	.984	.994	.812	.646	.726	.581	.981	.571	.522	.632
	VAE	.810	.969	.948	.992	.802	.576	.695	.491	.955	.478	.407	.579
Proposed	Baseline DAGMM	.093	.885	.802	.987	.513	.189	.105	.990	.554	.096	.055	.368
	S1-Student- t	.370	.325	.817	.203	.450	.122	.093	.178	.562	.114	.073	.263
	S1-Laplace	.807	.322	.809	.201	.641	.243	.186	.354	.775	.211	.158	.316
	S1-Skew-normal	.925	.060	.670	.032	.420	.139	.106	.202	.834	.262	.191	.421
	FlowDAGMM (S2)	<u>.998</u>	<u>.993</u>	<u>.994</u>	<u>.992</u>	<u>.817</u>	<u>.546</u>	<u>.538</u>	<u>.555</u>	<u>.933</u>	<u>.429</u>	<u>.391</u>	<u>.474</u>
	NGMF (S3)	.995	.989	.996	.982	.703	.333	.271	.433	.895	.244	.155	.579

**Figure 3: t-SNE visualisations of latent-space representations for each dataset.****Figure 4: Kernel density estimates of anomaly scores, showing separation between normal (blue) and anomalous (red) samples.**

the framework to high-dimensional image data; (3) attention-based autoencoders; and (4) more stable flow training objectives such as score matching.

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